EDA REPORT

Data-Driven Forecasting & Peak Demand Prediction

Afficionado Coffee Roasters

Exploratory Data Analysis • Demand Patterns • Peak-Hour Prediction • Business Insights • Power BI Readiness

1. Executive Summary

The Afficionado Coffee Roasters analysis covers 149,116 transaction records and develops a data-driven framework for understanding revenue, quantity, store performance, product-category contribution, intraday demand, and peak-demand prediction. The accompanying Power BI report provides operational views and an hourly demand-planning capability.

A key source limitation must be retained: the supplied transaction data contains a year and transaction time but does not contain a genuine calendar date. The notebook therefore constructs a timestamp from the available year and time. Consequently, the analysis supports intraday/hourly planning, but daily or weekly forecasting should not be presented as validated calendar-time forecasting.

Headline KPI	Value
Transactions	149,116
Quantity sold	214,470
Revenue	$698,812.33
Average transaction value	$4.69
Stores	3
Peak-demand model MAE	0.4365
Peak-demand model RMSE	0.4899

Key Business Findings

- Demand is broadly balanced across the three stores, so operational planning should remain store-specific.
- Coffee generates approximately 38.63% of revenue and Tea approximately 28.11%; together they represent about 66.74% of revenue.
- 10:00 is the strongest documented hour across all three stores.
- Hell's Kitchen records the highest documented peak-hour quantity at 9,873 units at 10:00.
- Lower Manhattan has the highest average transaction value at about $4.81.
- The current prediction model is suitable as a foundation for intraday peak-demand planning, subject to the source-data limitation.

2. Dataset Understanding & Data Quality

The raw Transactions sheet contains 149,116 rows and 11 columns. The notebook converts transaction identifiers, quantities, store/product identifiers and unit price to numeric types; standardizes transaction time; creates a transaction timestamp; derives hour, minute and daypart; calculates revenue; removes duplicate transaction IDs; and drops rows missing essential fields.

Engineered Field	Business Purpose
transaction_datetime	Time-based demand analysis
transaction_date	Calendar-date field derived from the constructed timestamp
hour / minute	Intraday demand and peak-hour analysis
daypart	Morning Rush, Lunch, Afternoon, Evening and Overnight segmentation
revenue	Transaction quantity × unit price
quantity_per_transaction	Basket-size proxy for operational planning
is_peak_hour	Store-specific flag based on the 75th percentile of hourly quantity

The notebook reports zero missing values across the analyzed fields after cleaning. However, missingness is not the main forecasting limitation: the absence of a genuine calendar date prevents reliable weekday, monthly, seasonal and year-over-year analysis.

3. Store-Level Business Analysis

Store	Transactions	Quantity	Revenue	Avg. Transaction
Hell's Kitchen	50,735	71,737	$236,511.17	$4.66
Astoria	50,599	70,991	$232,243.91	$4.59
Lower Manhattan	47,782	71,742	$230,057.25	$4.81

- Hell's Kitchen leads the three stores in revenue.
- Lower Manhattan has the highest average transaction value despite having the fewest transactions.
- Quantity is nearly evenly distributed across stores, with each contributing roughly one-third of total quantity.
- Staffing and inventory decisions should therefore be based on each store's hourly demand pattern rather than a single network-wide average.

4. Product Category Analysis

Category	Transactions	Quantity	Revenue	Revenue Share
Coffee	58,416	89,250	$269,952.45	38.63%
Tea	45,449	69,737	$196,405.95	28.11%
Bakery	22,796	23,214	$82,315.64	11.78%
Drinking Chocolate	11,468	17,457	$72,416.00	10.36%
Coffee beans	1,753	1,828	$40,085.25	5.74%
Branded	747	776	$13,607.00	1.95%
Loose Tea	1,210	1,210	$11,213.60	1.60%
Flavours	6,790	10,511	$8,408.80	1.20%
Packaged Chocolate	487	487	$4,407.64	0.63%

- Coffee is the dominant category, generating $269,952.45 and approximately 38.63% of revenue.
- Tea contributes $196,405.95 and approximately 28.11% of revenue.
- Coffee and Tea together account for approximately 66.74% of revenue, making them central to peak-period inventory and staffing decisions.
- Bakery and Drinking Chocolate are the next meaningful contributors, while several smaller categories contribute relatively little revenue.

5. Peak Demand & Intraday Pattern Analysis

The notebook aggregates transactions by store, hour and daypart and defines a store's peak hour when hourly quantity is at or above that store's 75th percentile. The resulting peak table shows a strong concentration around the morning operating window.

Store	Documented Peak Hours	Highest Documented Quantity
Astoria	07:00–10:00	7,388 at 10:00
Hell's Kitchen	08:00–11:00	9,873 at 10:00
Lower Manhattan	07:00–10:00	9,452 at 10:00

- 10:00 is the strongest documented peak across all three stores.
- Hell's Kitchen reaches the largest documented peak-hour quantity, 9,873 units at 10:00.
- Astoria and Lower Manhattan show a 07:00–10:00 morning-rush pattern; Hell's Kitchen remains in the peak set through 11:00.
- The percentile-based peak flag adapts to each store's own demand distribution.

6. Peak-Demand Prediction Model

The notebook uses a HistGradientBoostingRegressor with a preprocessing pipeline. Categorical variables are one-hot encoded, while unit price, hour and minute are numeric features. The data is ordered by hour and minute and split using an 80%/20% sequential cutoff.

Model Component	Reported Result
Model	HistGradientBoostingRegressor
MAE	0.4365
RMSE	0.4899
Categorical features	Store location, product category, product type
Numeric features	Unit price, hour, minute
Train/Test split	80% / 20% sequential split

Interpretation

The notebook reports relatively small errors for the transaction-level quantity target under its test setup. These metrics should not be interpreted as evidence of a validated long-horizon forecast. The model uses intraday features and categorical context rather than lagged daily demand with genuine calendar dates.

Model Improvement Opportunities

- Add a genuine transaction_date field.
- Introduce weekday, month, holiday and seasonal features.
- Add lagged demand and rolling-average features once multiple dates are available.
- Use rolling or expanding time-series validation instead of a simple sequential cutoff.
- Evaluate forecasts on genuinely future periods.

7. Power BI Dashboard & Decision Use

Dashboard Area	Decision Question
Executive Overview	How much revenue, quantity and transaction activity are being generated?
Peak Demand & Operations	When are stores busiest and where should operational resources be concentrated?
Product & Store Analysis	Which categories and stores drive sales and volume?
Hourly Demand Forecast & Planning	What does the near-term hourly pattern suggest for planning?

The supplied PBIX also contains report sections named Home, Exclusive Overview, PEAK DEMAND & OPERATIONS, Product & Store Analysis, and Hourly Demand Forecast & Planning. Its forecast visual uses an hourly datetime field and forecast transformation. This should be described as hourly planning/forecasting support until the calendar-date limitation is resolved.

8. Consolidated Business Insights

- Peak staffing: concentrate staffing, barista capacity and service preparation around 07:00–10:00, with Hell's Kitchen requiring attention through 11:00.
- Inventory readiness: protect Coffee and Tea availability before the morning rush because these categories represent about 66.74% of revenue.
- Store-specific planning: overall demand is balanced, but peak-hour curves differ by store.
- Revenue opportunity: Lower Manhattan has the highest average transaction value, so basket-building initiatives can be evaluated there without assuming it is the highest-volume store.
- Operational monitoring: the peak flag and hourly demand table provide a repeatable way to identify likely capacity constraints.
- Forecast governance: use the current model for intraday peak-demand prediction, while treating daily/weekly forecasting as a future enhancement requiring true transaction dates.

9. Recommended Actions

Priority	Action	Expected Business Use
1	Prepare inventory before the morning peak	Reduce stockout risk during highest-demand hours
2	Schedule staffing by store and hour	Align labor with actual intraday demand
3	Protect Coffee and Tea availability	Focus replenishment on revenue-driving categories
4	Monitor 10:00 demand separately	Manage the common highest-volume hour
5	Add a true transaction_date field	Enable valid daily/weekly seasonality and forecasting
6	Back-test future forecasts	Measure performance on genuinely future periods

10. Limitations & Next-Step Improvements

- The source notebook explicitly states that the supplied data lacks a true calendar date. The constructed transaction_datetime should not be treated as a genuine daily timeline.
- The model evaluation uses an 80/20 sequential split after sorting by hour and minute. Stronger forecasting evaluation should preserve chronological order using actual dates and rolling/expanding validation.
- The current peak-demand model does not use lagged demand, rolling averages, holidays, weekdays or weather.
- The PBIX hourly forecast should be labeled as planning/forecasting support rather than a validated daily demand forecast until the date limitation is resolved.
- Business impact measures such as stockout reduction, labor savings, service time, waste reduction or incremental revenue are not provided by the supplied notebook and should be measured after deployment rather than invented.

11. Conclusion

The Afficionado Coffee Roasters analysis establishes a data-driven framework for understanding transaction demand and managing peak operating periods. The dataset contains 149,116 clean transactions, $698,812.33 in revenue and 214,470 units. Demand is broadly balanced across the three stores but becomes concentrated during the morning rush, with 10:00 emerging as the strongest documented hour across all stores. Coffee and Tea are the major revenue drivers. The peak-demand model reports MAE 0.4365 and RMSE 0.4899 under the notebook's test setup, providing a useful foundation for intraday operational planning.

The most important next step is to add a genuine transaction date. That enhancement would allow robust daily/weekly forecasting, weekday and seasonal effects, lag/rolling features, proper time-series backtesting, and stronger peak-demand planning. Until then, the project is best positioned as an hourly demand and peak-demand prediction system with Power BI operational planning.